

CSCI 264 — Assembly Language

Lab 4

Strings and Structs in C

Start from `lecture_10.c`, the roster program we wrote together in class on Friday. Copy it into your lab directory and rename it `roster.c`. It already defines `struct studentT`, allocates an array of five students on the heap, fills them in with `init_student`, and prints them.

You are adding two functions to that program and a few lines to `main`. Do not change the struct definition or `init_student`.

Functions

You must write each of these functions:

```
struct studentT *find(struct studentT *roster, int n, char *name);
void print_layout(struct studentT *s);
```

- `find` searches the roster for the student whose `name` field matches `name`. It returns a pointer to that student, or `NULL` if nobody on the roster has that name. Compare the names with `strcmp`: `==` would compare the two addresses rather than the characters at them. Remember that `strcmp` returns 0 when the strings *match*.
- `print_layout` takes a pointer to one student and prints the address of the struct itself, then the address of each of its four fields together with that field's *offset* — the number of bytes from the start of the struct to the start of the field.

Offsets

An offset is a subtraction, not something you count by hand. Cast both addresses to `char *` before subtracting, so that the difference comes out in bytes:

```
(char *)&s->age - (char *)s
```

The casts are the whole trick. Subtracting two `struct studentT *` values would give you a number of structs; subtracting two `int *` values would give you a number of ints. A `char` is one byte, so casting both sides to `char *` is how you ask for the answer in bytes. Print the result with `%ld`.

For the `name` field, use `s->name` as the address: `name` is an array, so it already evaluates to the address of its first element.

In main

After the printing loop that is already there, add the following:

1. Call `find` twice — once with a name that is on the roster, and once with a name that is not. Print the student you found in the first case and a message in the second. Your program must not crash on the second call.
2. Call `print_layout` on the student that `find` returned.
3. Print `sizeof(struct studentT)` using `%lu`.
4. Print `roster`, `roster + 1`, and `roster + 2`.

Output

A run of your program should look like this, with real addresses in place of `0x...`:

```
... the five students, printed by the loop already in the program ...
```

```
found Ruth Teitelbaum age 21 gpa 3.2 class of 2027
Ada Lovelace is not on the roster
```

```
[layout] s           = 0x...
[layout] &s->name     = 0x... offset 0
[layout] &s->age      = 0x... offset 64
[layout] &s->gpa      = 0x... offset 68
[layout] &s->grad_yr  = 0x... offset 72
```

```
[main]  sizeof(struct studentT) = 76
[main]  roster      = 0x...
[main]  roster + 1 = 0x...
[main]  roster + 2 = 0x...
```

The four offsets are not addresses and will be the same on every machine in the lab. If yours are not 0, 64, 68, and 72, something is wrong with your subtraction.

Questions

Answer these in a comment block at the bottom of `roster.c`. Two or three sentences each is plenty.

1. How many bytes apart are `roster` and `roster + 1`? You added 1 to a pointer — where did that number come from?
2. Add `char class_yr;` as a fifth field at the end of `struct studentT` and recompile. What does `sizeof` report now? You added one byte to the struct; how many bytes did it grow by? Submit the version *without* the extra field.
3. `init_student` sets `s->name[NAME_LEN - 1] = '\0'`; after calling `strncpy`. Suppose that line were deleted and a student were initialized with a 70-character name. What would `printf("%s", s->name)` do, and which field is standing in its way?

Submission

Submit `roster.c` on Canvas and demo the program to Professor West before you leave lab. **If you do not finish during the period, submit on Canvas by Friday, September 25 and demo during office hours.** Put your name in a comment at the top of the source file. If a classmate helped you put their name in the comment. Disclose any use of Generative AI in a comment at the top.